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Department of Child Health

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Title: Sapropterin Dihydrochloride Protocol – (KUVAN)

Introduction:

Sapropterin Dihydrochloride (Kuvan) is an orally **active synthetic form** of (6R)-L-erythro-5,6,7,8-**tetrahydrobiopterin (BH4)**; a naturally occurring enzyme cofactor that works in conjunction with phenylalanine hydroxylase (PAH) to metabolize Phe, Tyrosine hydroxylase (TH) and Tryptohane Hydrpxylase (TH) that required for the synthesis of serotonin, dopamine and nitric oxide.

Kuvan has received orphan drug designation from both the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA) In November 2007 as an oral treatment for patients suffering from HPA due to PKU or BH4 deficiency.

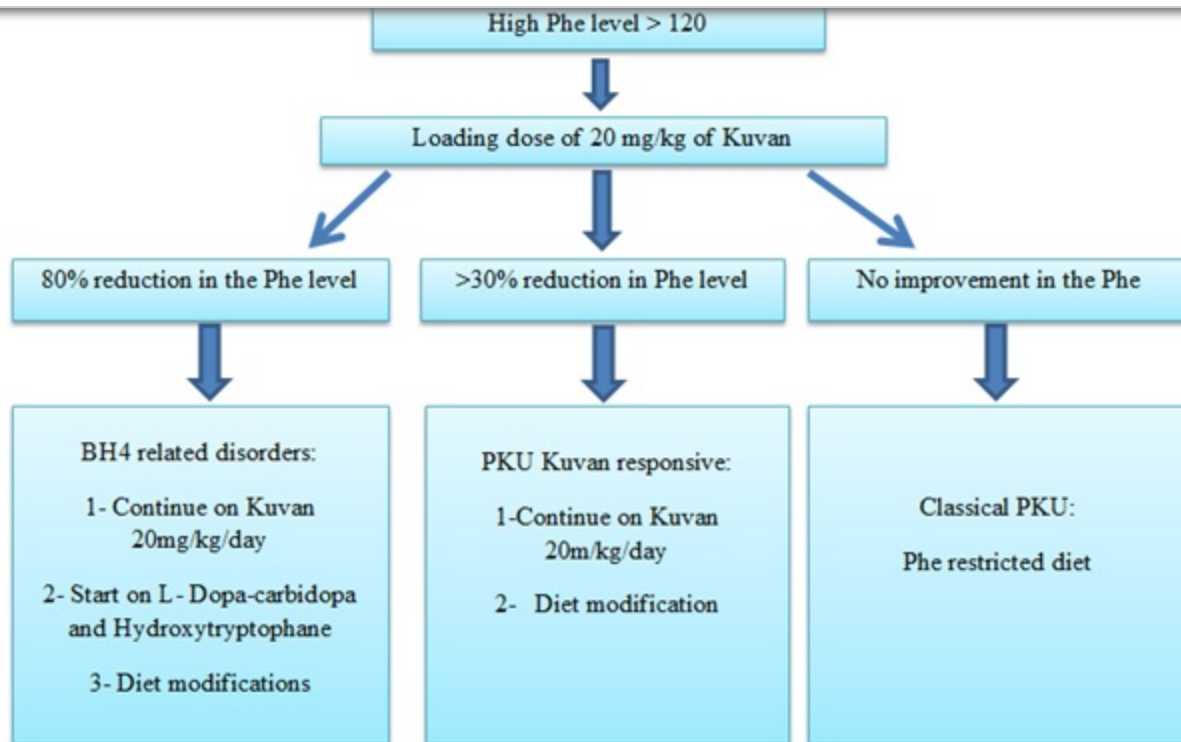
Indication:

- Hyperphenylalanimeia
- Mild-to-moderate PKU

About PKU

Phenylketonuria occurs due to an inherited deficiency of the enzyme phenylalanine hydroxylase. If untreated it causes an excessive accumulation of the amino acid phenylalanine in the body which prevents normal brain development. The established treatment for phenylketonuria consists of dietary restriction of natural protein but with prescribed phenylalanine-free amino acid, mineral and vitamin supplements. With this treatment the long-term outcome for people with phenylketonuria is excellent but the diet is onerous. Sapropterin dihydrochloride, the cofactor for phenylalanine hydroxylase, could lower phenylalanine concentration significantly in phenylketonuria and might allow a relaxation of dietary restrictions.

About BH4 deficiency related disorders: as mentioned earlier Tetrahydrobiopterin (BH4) is a pivotal enzymatic cofactor required for the synthesis of serotonin, dopamine and nitric oxide. BH4 is essential for numerous physiological processes at periphery and central levels certainly for neurotransmitter. Deficiency in the enzyme involve in the synthesis of BH4 results in wide range of symptoms from very mild to severe. Babies with BH4 deficiency appear normal at birth but may develop neurological symptoms such as abnormal muscle tone; poor head control, seizures, and delayed motor development. Without treatment, the condition can cause permanent intellectual disability. The Maine state of treatment is to replace the cofactor and the deficient neurotransmitters products.



Monitoring & Follow up:

1. Phe level to monitors monthly in newly diagnosed patient and then every 3 months once the target Phe level is achieved.
2. The dose can be reduced to 10 mg/kg/day providing there is no elevation in the level.
3. The treatment will be life long for the patient with BH4 related disorders and PKU patient.

References:

1. Sapropterin dihydrochloride for Sapropterin dihydrochloride for phenylketonuria, Usha Rani Somaraju, Marcus Merrin, and Cochrane Cystic Fibrosis and Genetic Disorders Group, 2015.
2. Response to Sapropterin hydrochloride.

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